

Continuous Concurrent Feedback Degrades Skill Learning: Implications for Training and Simulation

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In two experiments we investigated the role of continuous concurrent visual feedback in the learning of discrete movement tasks. During practice the learner's actions either were or were not displayed on-line during the action; in both conditions the participant received kinematic feedback about errors afterward. Learning was evaluated in retention tests on the following day. We separated (a) errors in the fundamental spatial-temporal pattern controlled by the generalized motor program from (b) errors in scaling controlled by parameterization processes. During practice concurrent feedback improved parameterization but tended to decrease program stability. Based on retention tests, earlier practice with continuous feedback generally interfered with the learning of an accurate motor program and reduced the stability of time parameterization. Continuous feedback during acquisition degrades the learning of not only closed-loop processes in slower movements (as has been found in earlier studies) but also motor programs and their parameterization in more rapid tasks. Implications for feedback in training and simulation are discussed.

INTRODUCTION

Over the past several decades, researchers have examined the role of numerous types of augmented (extrinsic, or supplemental) information feedback for the learning of movement skills. (For reviews, see Magill, 1993; Salmoni, Schmidt, & Walter, 1984; Schmidt, 1991.) One type of augmented feedback that has received both practical and theoretical attention in training and simulation settings is termed *concurrent feedback*: supplementary information presented to the learner during the actual action. This feedback

can be presented discontinuously to signal that the performer is on target at that moment or that a certain level of performance is being achieved. This feedback can also be presented continuously (on-line, with essentially no delay) to indicate the level of momentary performance error (Karlin & Mortimer, 1963; Kohl & Shea, 1995; Phillips & Berkhout, 1978), deviations from a goal movement pattern (Van der Linden, Cauraugh, & Greene, 1993), the pattern of ongoing electromyographic activity (Mulder & Hulstijn, 1985), or other sources of biofeedback. Continuous concurrent augmented feedback using vision is the focus of the present article.

Variants of this type of feedback are used in many practical situations of interest to the

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human factors/ergonomics field, such as simulators, in which a feature to be controlled (e.g., the location of an aircraft with respect to a glide path during landing; Lintern, 1980) is fed back to the learner concurrently and continuously with the action. In physical therapy, continuous feedback has been used for decades as a way to train weight-bearing in injured limbs (Winstein, 1991).

On the surface, continuous feedback appears to be effective for learning because it guides the learner powerfully to the correct response, minimizes errors, and holds behavior on target. The problem is that the performance gains during practice are seldom carried over to retention or transfer tests in which the augmented feedback is withdrawn. The usual finding is that people who have practiced with concurrent continuous feedback often perform worse on no-feedback retention tests than do people who have practiced without such feedback. In other words, continuous concurrent feedback appears to enhance performance during practice when the feedback is operating, but it does not contribute to learning and may even degrade learning, as measured on retention and transfer tests (see Annett, 1959, 1969; Karlin & Mortimer, 1963; Kohl & Shea, 1995; Patrick & Mutlusoy, 1982; Van der Linden et al., 1993).

If augmented continuous feedback generally degrades learning, it should be avoided in the design of learning settings. Nevertheless, augmented continuous feedback has been deliberately included in both simulators and training programs, often with impressive realism and fidelity but usually at considerable expense. Its inclusion is understandable in that it does generally improve performance when it is present, and it seems obvious at first glance that such information must contribute to learning as well. However, recent views of training (e.g., Schmidt & Bjork, 1992) emphasize that measures of performance during training are not generally good predictors of longer-term learning, which is usually best evaluated on retention or transfer tests that are separated from the training setting. Even more critical, there is considerable evidence that, relative to some so-called standard practice con-

dition, several factors that facilitate performance during practice are detrimental to retention and/or transfer performance, which is widely considered important in the success of training and simulation programs.

Hypotheses for Feedback Processing

The puzzling and counterintuitive failure of continuous concurrent feedback to contribute to learning (even though it contributes strongly to performance during practice) has been explained mainly by the notion that concurrent feedback is overly guiding during practice. Originally suggested by Annett (1959, 1969), this idea has been formalized recently to form what is termed the *guidance hypothesis* for feedback (Salmoni et al., 1984; Schmidt, 1991; Schmidt, Young, Swinnen, & Shapiro, 1989). In this view the learner uses the guidance of concurrent feedback as a crutch to produce the correct action during practice. However, this reliance on feedback prevents the learner from exercising the capabilities that are needed on the retention or transfer test, on which augmented feedback is typically not available, so performance falls below even that of learners who did not have continuous feedback during practice. These overly guiding properties of concurrent feedback can be thought of in two ways.

Guidance properties. Concurrent feedback seems to be most commonly used to guide behavior. This process is probably most apparent in tasks for which the movement duration is relatively long (greater than several seconds). Most writers would agree that in such tasks, feedback processing is a critical part of the action. The augmented concurrent feedback indicates the nature and direction of needed moment-to-moment corrections and provides additional ways in which the information can be used for action (perhaps more directly than intrinsic feedback), facilitating momentary performance. Tasks in which this seems common are slow lever-positioning responses (Fox & Levy, 1969; Kohl & Shea, 1995), movement-patterning tasks (Van der Linden et al., 1993), and complex pursuit or compensatory tracking activities. In such tasks

open-loop programming processes seem to be of minimal importance to the movement's control.

Attentional processes. A second, related view is that concurrent feedback during practice distracts attention from the intrinsic feedback that is naturally present. Attending to these augmented sources prevents the learner from acquiring the capability to deal with the intrinsic information that will be present in the retention test and on which his or her performance will presumably depend when the augmented feedback is withdrawn. Most, if not all, experiments on continuous concurrent feedback have used such tasks, and their results can usually be interpreted in this general way.

A surprising counterexample is from Swinnen, Lee, Verschueren, and Serrien (in press), in which concurrent feedback facilitated both the performance and learning of a relatively difficult, long-duration bimanual coordination pattern. Lintern (1980) also showed some benefits of concurrent feedback in an aircraft simulator, in which a simple ("skeletal") display depicting a runway approach was augmented in various ways (e.g., the desired flight path information). The reasons for these contradictory patterns of results are unclear, and further work seems warranted to understand the nature of these effects.

Feedback in Motor Program Learning

An omission in the human factors literature concerns tasks for which open-loop programming processes are strongly involved—chiefly tasks that are performed relatively rapidly in stable, predictable environments (e.g., hammering, tossing, rapid control adjustments; see Schmidt, 1988). In such situations the preprogramming of an action would have the largest possibility of being successful, as environmental variations necessitating feedback-based modifications would be minimal or absent. Most researchers agree that such tasks depend little, if at all, on feedback processes. Therefore, if concurrent continuous feedback were presented during practice of these actions, would practice performance be enhanced and retention performance be degraded, as it is in tasks for which feedback

processing is important? Alternatively, because these tasks are thought to be performed largely on the basis of programming control, would these degrading effects of concurrent feedback be absent for this class of tasks, or would this feedback even facilitate learning?

To our knowledge there have been no previous attempts to evaluate the role of concurrent feedback on rapid, programmed activities; this formed a major goal of the present experiment. As such actions are frequently involved in many real-world training tasks, such information should be of value to trainers in the human factors field.

The additional goal of these experiments was to evaluate the nature of the effects, if any, of concurrent continuous feedback on the acquisition of several component processes of actions. In our previous work (Wulf & Schmidt, 1994; Wulf, Schmidt, & Deubel, 1993) we used a task and paradigm in which the learners acquire novel, relatively brief (<1 s) movement patterns (see also Wulf, Lee, & Schmidt, 1994, for similar procedures). In such tasks we have been able to separate the capability to produce an effective action into two components. One part involves the fundamental spatial-temporal patterning (or basic form) of the action and, in our terms, is based on the acquisition of a generalized motor program (GMP). The other part involves processes responsible for scaling the action properly in order to meet various environmental demands (e.g., size and speed) or, in our terms, the capability to parameterize the GMP.

Earlier we presented evidence that various experimental factors contribute differentially to these processes, adding to our confidence that these components of responding are psychologically and behaviorally distinct. Thus our second goal was to examine whether augmented concurrent feedback influences the acquisition of a GMP and/or the capability to parameterize the action to meet environmental demands.

EXPERIMENT 1

In Experiment 1 we examined the role of concurrent augmented visual feedback in learning a

relatively complex arm action. Concurrent feedback was either provided or absent, and learning was evaluated on a delayed retention test without such augmented feedback.

Method

Participants. Students at the University of Munich ($N = 30$) participated in exchange for DM 15 (about US\$9.00) for their services. The participants had no prior experience with the task, and they were not informed about the purposes of the experiment.

Apparatus

The apparatus consisted of a horizontal lever attached at one end to a vertical axle that turned almost frictionlessly in ball bearing supports. The supports were mounted to the side of a table, allowing the lever to move in the horizontal plane over the table surface. A vertical handle was attached at the other end of the lever; its position could be adjusted so that when the student's forearm rested on the lever, the elbow was aligned over the axis of rotation while the student grasped the handle comfortably. A potentiometer was attached to the lower end of the axle to record position; its output was sampled at 200 Hz by a Hewlett Packard Vectra QS/20 computer. A wooden cover over the tabletop prevented the seated participant from seeing the lever.

Task

Participants were asked to produce a right-arm lever movement with certain spatiotemporal goal movement patterns. Four goal movement patterns were used in the experiment, each with the same fundamental pattern (i.e., with the same relative timing and relative amplitudes). These figures were linearly scaled to different absolute amplitudes to create the four task versions, but the rate scaling was constant to produce a duration of 937 ms for each version (see Figure 1).

Before each trial the goal movement pattern was displayed for 1.5 s on a computer screen (EIZO Flexscan 9060S). A letter (*A*, *B*, or *C*) denoting a specific pattern was displayed in the upper left corner of the screen, together with the

goal movement pattern, to facilitate discrimination between the different patterns. The goal pattern and letter were then removed and replaced by two vertical cursors, one in the center representing the starting position and another representing the lever's present position. The participant then moved the lever to the starting position, indicated by the alignment of the two cursors. The two cursors then disappeared; immediately following was a tone indicating that the movement could begin. The participant's task was to make a lever movement that corresponded in space and time to the goal pattern presented earlier. Once the participant began moving the lever, the movement was recorded for 1.6 s.

After a 3-s interval, knowledge of results (KR) was provided for 5 s by superimposing the spatiotemporal trace of the participant's movement over that of the goal pattern. The goal pattern was displayed in white (on a black background); the participant-produced pattern was in yellow. In the lower right corner, *Vorgabe* (template) and *Reproduktion* (reproduction) were written in the same colors as the respective traces. Also, as the movement was recorded for 1.6 s, the subject's trace extended farther to the right than the goal pattern, allowing the two curves to be distinguished easily. A perfect movement was indicated by exact superimposition of the goal trace and the participant's trace (until the end of the goal trace), and deviations between the two traces indicated errors.

In addition to the two traces, the root-mean-square error (RMSE) between the participant's movement and the goal movement (for the duration of the goal pattern) was displayed in the upper right corner on the screen. Also, the trial number of the just-performed movement appeared in the upper left corner of the screen.

Procedures and Experimental Conditions

Participants were randomly assigned to one of two groups: one that received concurrent feedback (ConFB) and one that did not (No-ConFB). Both groups received KR (as described in the previous section) after each practice trial, but in addition the ConFB group received concurrent

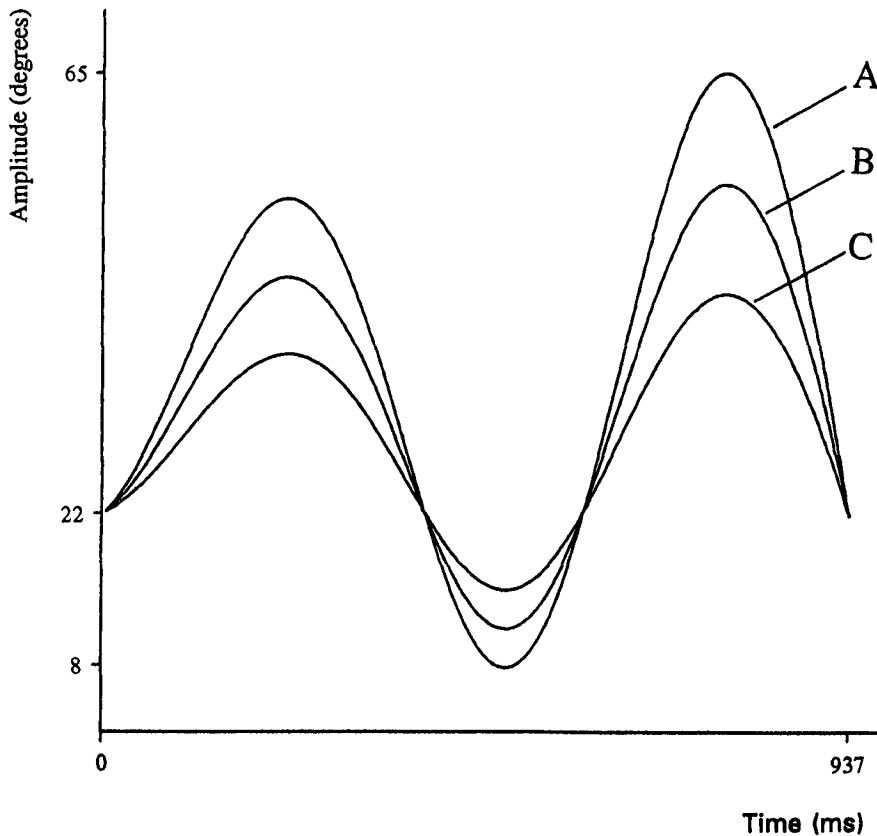


Figure 1. Spatiotemporal functions of the goal movement patterns.

feedback; that is, they saw their trace being drawn on the screen in real time as they produced the movement. For the ConFB group, once the lever was in the start position, the goal pattern appeared on the screen. As soon as the ConFB participant began moving the lever from the start position, the pattern he or she produced was drawn in real time over the goal pattern throughout the next 1.6 s, after which the screen turned blank. (For the No-ConFB group, the screen remained blank during the movement.) After 3 s, KR was provided to both groups.

Thus the only difference between the No-ConFB and the ConFB groups was that the latter was shown the position-time curve of their movement in relation to that of the goal pattern while the movement was being produced, whereas the former did not receive this continuous feedback during the movement. Note that both groups re-

ceived identical postmovement feedback information.

After the procedures had been explained and before the practice phase began, participants performed three practice trials under their respective practice conditions using Pattern B (see Figure 1). Their performance was discussed after each trial to ensure that they understood how to interpret the feedback. All participants then performed 90 practice trials using Patterns A, B, and C (see Figure 1) presented in random order, with the restriction that each pattern appeared 30 times.

One day after the practice phase, all participants performed a retention test consisting of 12 trials without either concurrent feedback or KR; the three practice-task versions (A, B, and C) were presented in random order (four trials per version). Note that the concurrent feedback that

had been received by the ConFB group was now removed, so that the retention conditions were identical across groups.

Statistical Analyses and Dependent Measures

The analysis and computation of dependent measures focused on separating the errors in the underlying pattern (based on the GMP) from the errors in scaling the pattern (based on parameterization processes). These methods have been described previously (e.g., Wulf & Schmidt, 1994; Wulf et al., 1993), and we will not repeat the full description here. However, a brief summary of this method, with the interpretation of the various scores it generates, is given next.

Root-mean-squared error (RMSE). First, as a measure of overall error, we calculated the root-mean-squared deviations between the participant's pattern and the template across the 937 ms of the template. This measure is sensitive to both constant errors (the average deviation of the participant's movements from the goal pattern) and within-subject variability (Schmidt, 1988). It is important to note that RMSE is sensitive to both errors in GMP production and errors in parameterization, which are separated using the measures described next.

Scaling factors. Each participant-produced trajectory and its goal pattern were first synchronized at the moment that a displacement of 0.65° from the baseline was detected. Next, a computer program linearly scaled (stretched or compressed) the participant-produced trajectory in time by various factors. For each temporal rescaling factor, the within-subject correlation between the participant's rescaled trajectory and the goal pattern was calculated. This process was repeated with scaling factors ranging from 0.2 to 2.0 times the actual rate, which easily spanned the range of overall movement duration errors. We then selected the scaling factor for which the correlation with the goal pattern was maximized. This process did not introduce phase shifts between the two functions, only proportional changes in rate, with the start of the movement fixed.

The scaling rate that yielded the maximum cor-

relation between the participant's trace and the goal pattern was taken as a temporal scaling factor, or time factor, indicating the error in temporal parameterization. Thus the temporal scaling factor indicated the amount by which the participant's trajectory had to be linearly expanded or compressed in time so that it correlated maximally with the goal pattern. A correct temporal scaling had a value of 1.0, and values larger (or smaller) than 1.0 indicated movements with rates that were too slow (or fast). These were expressed as absolute constant error (|CE|) in time scaling for each participant (the absolute value of the mean across trials, indicating systematic bias) and as variable error (VE) in time scaling (the standard deviation [SD] across trials, indicating variability).

Next, an amplitude factor was determined as the variance in amplitudes (computed over the 193 samples in a trial) of the optimally time-scaled movement trajectory divided by the variance of the goal pattern. This ratio was taken as a measure of error in amplitude parameterization and indicated the amount by which the subject's trajectory had to be linearly scaled in amplitude so that it fit the goal pattern optimally. Again, the correct value for amplitude scaling was 1.0, and values larger (or smaller) than 1.0 indicated amplitudes that were too large (or small). These were expressed as |CE| and VE in a manner analogous to the temporal scaling factors.

Residual RMSE. For each trial the remaining (residual) RMSE between the optimally time- and amplitude-scaled trajectory and the goal pattern was calculated. We interpreted this residual RMSE as the inaccuracy of the GMP, as it represented the disagreement of the movement and the goal pattern when the errors in amplitude and temporal parameterization had been removed. Two measures were derived from the residual RMSE calculations.

First, the mean residual RMSE over a series of trials gives an estimate of the deviation in the participant's average trajectory from the goal pattern. It is mathematically equivalent to measuring the average movement trajectory for a series of trials and then computing the RMS deviation

between this average trajectory and the goal pattern. It represents the deviation of the participant's average, or typical, trajectory from the goal pattern and is taken as a measure of GMP accuracy.

Second, residual RMSE-VE (a variable error) was calculated as follows. Within a block of trials, and using the rescaled templates, the within-subject (over trials) *SDs* of the amplitudes of the first through the last samples were computed (193 samples/movement were used). This provided estimates of the amplitude variability of the participant-produced templates. The mean of these 193 separate estimates provided a kind of average variability of the GMP's amplitude and was the residual RMSE-VE reported here.

As with many global performance scores, such a measure is insensitive to the fact that some portions of the movement are inherently more or less variable than others, and it contains several additional sources of variability (recording errors, neuromuscular variability); as a result, this measure will not be a very pure estimate of the extent to which the GMP is unstable. Within these limitations, we interpreted it as a measure that was at least sensitive to the instability of the amplitude characteristics of the GMP.

Interpretations. These scaling processes may be more clearly understood from Figure 2. The solid trace in each panel represents the goal movement pattern, parameterized properly. The participant's movement trajectory is indicated by the dashed trace. The dotted trace is the participant's movement rescaled in time and amplitude via the procedures described earlier. The deviation between the dotted trace and the goal pattern was the basis for the residual RMSE. In the upper panel the residual RMSE is relatively small after rescaling, indicating a fairly accurate GMP; the lower panel shows a much larger residual, indicative of a less accurate GMP.

Results: Practice

Overall performance. As can be seen from Table 1, both groups reduced their RMSEs over the course of practice. Throughout the entire prac-

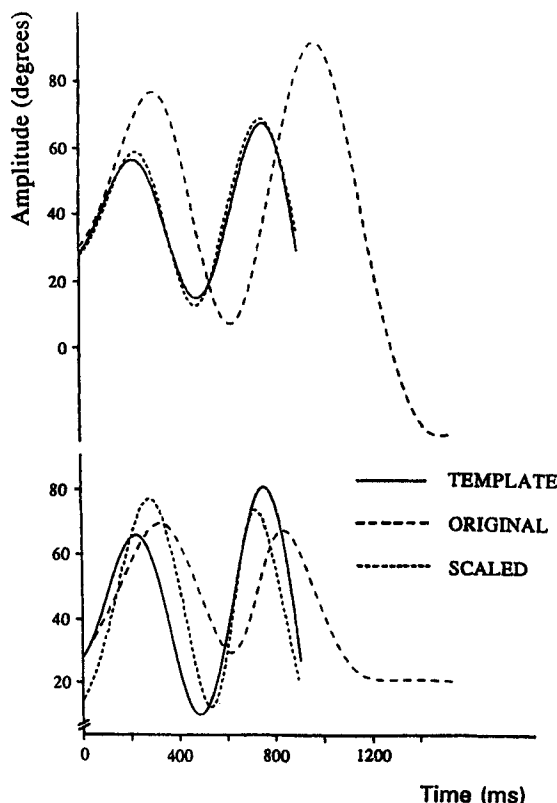


Figure 2. Examples of the effect of scaling movement trajectories; the solid trace is the goal movement pattern, the dashed trace is the participant's trajectory, and the dotted trace is the scaled version of the participant's trace.

tice phase, the ConFB group showed a greater overall accuracy in producing the goal pattern than did the No-ConFB group; continuous feedback was clearly effective in facilitating performance. The main effects of both block, $F(5, 135) = 31.5$, $p < .001$, and group, $F(1, 27) = 12.4$, $p < .01$, were significant. There was no Group $\times$ Block interaction, $F(5, 135) < 1$. (The practice data of one participant in the No-ConFB group were accidentally deleted.)

Motor-program measures. Contrary to the global performance measure (RMSE), there were essentially no differences between groups during practice with regard to residual RMSE (Figure 3, left). The ConFB and the No-ConFB groups *seemed* similarly effective in producing the fundamental movement pattern, or GMP. The main

TABLE 1

Mean RMSE, Amplitude Factor |CE|, Amplitude Factor VE, Time Factor |CE|, and Time Factor VE in Practice and Retention in Experiment 1

	Practice Blocks						Retention
	1	2	3	4	5	6	
RMSE (overall performance)							
ConFB	296	242	209	189	178	162	380
No-ConFB	404	303	321	265	274	246	359
Amplitude Factor ICEI (amplitude parameter accuracy)							
ConFB	.10	.10	.10	.11	.09	.08	.42
No-ConFB	.30	.40	.35	.21	.19	.16	.32
Amplitude Factor VE (amplitude parameter stability)							
ConFB	.30	.26	.23	.21	.19	.18	.27
No-ConFB	.57	.76	.57	.19	.19	.23	.25
Time Factor ICEI (time parameter accuracy)							
ConFB	.09	.08	.04	.03	.03	.02	.20
No-ConFB	.18	.10	.11	.08	.08	.07	.14
Time Factor VE (time parameter stability)							
ConFB	.16	.12	.08	.08	.07	.06	.16
No-ConFB	.25	.18	.17	.13	.13	.12	.10

effect of block was significant, $F(5, 135) = 23.0$, $p < .001$, indicating that both groups reduced their errors with practice. The group main effect, $F(1, 27) < 1$, and the interaction of group and block, $F(5, 135) = 1.1$, $p > .05$, were not significant.

Participants also reduced their residual RMSE-

VEs across practice blocks (Figure 4, left), indicating that they generally became more consistent in producing the fundamental movement structure. However, the ConFB group was clearly more variable in the fundamental pattern than the No-ConFB group. The main effects of both

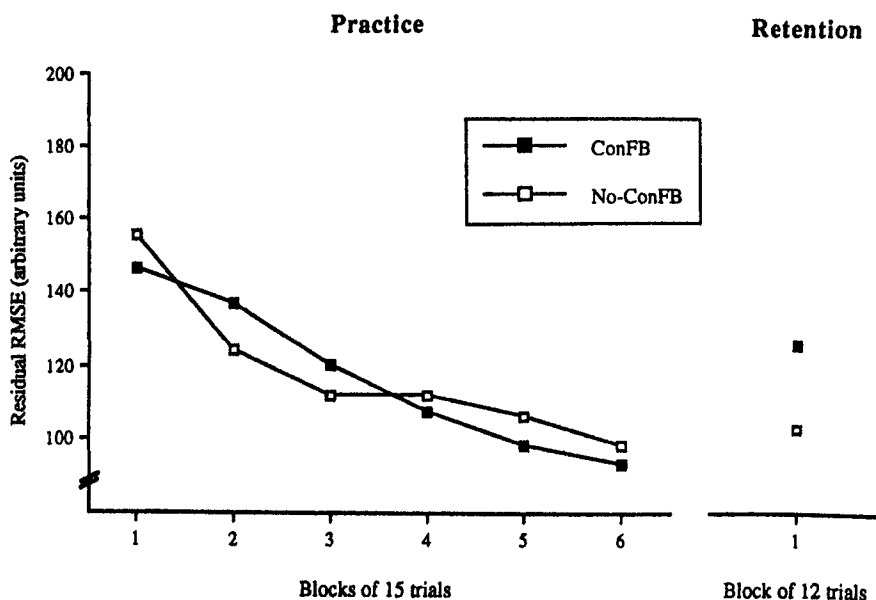


Figure 3. Residual RMSE in practice and retention in Experiment 1.

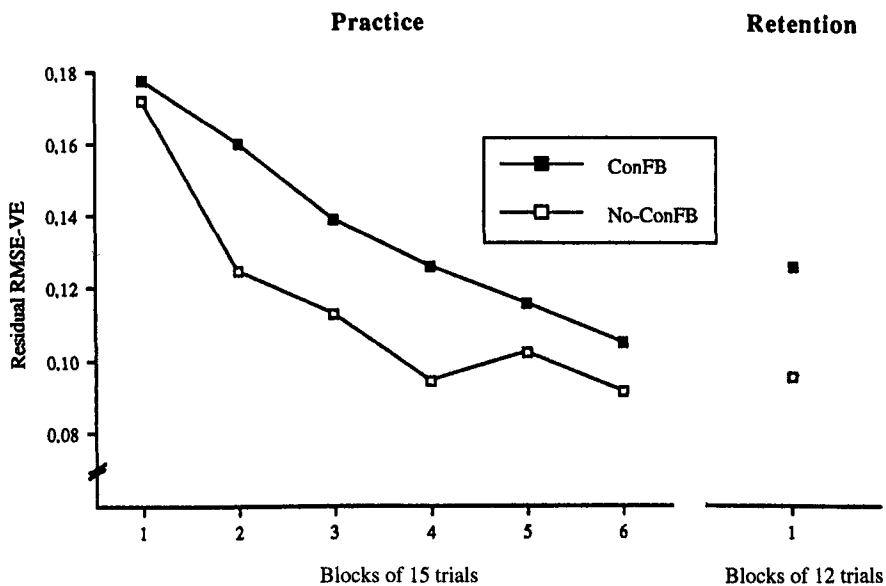


Figure 4. Residual RMSE-VE in practice and retention in Experiment 1.

block, $F(5, 135) = 26.9$, $p < .001$, and group, $F(1, 27) = 4.3$, $p < .05$, were significant. The Group $\times$ Block interaction was not significant, $F(5, 135) = 1.1$, $p > .05$.

Parameterization measures. Amplitude factor ICEIs are shown in Table 1. The No-ConFB group had comparatively large ICEIs in amplitude factors in the first half of practice and smaller errors in the second half, whereas the ConFB group was far more accurate in parameterization throughout practice. Thus continuous feedback clearly facilitated parameterization accuracy, relative to the KR-only condition. The main effect of group was significant, $F(1, 27) = 8.6$, $p < .01$, but the block main effect, $F(3, 135) = 2.1$, $p = .07$, and the interaction of group and block, $F(3, 135) = 1.7$, $p = .15$, did not quite reach conventional levels of significance.

For amplitude factor VEs, the ConFB group demonstrated relatively small and stable values throughout practice, whereas the No-ConFB group produced larger VEs during the first half of practice, with reduced errors to the level of the ConFB group in the second half (see Table 1). However, the main effect of group, $F(1, 27) = 3.5$, and block, $F(5, 135) = 1.7$, and the Group $\times$ Block

interaction, $F(5, 135) < 1$, were nonsignificant, all $ps > .05$.

For time parameterization, both groups became more accurate over practice, with the ConFB group demonstrating clearly smaller time factor ICEIs than the No-ConFB group throughout the whole practice phase (see Table 1). The main effect of group, $F(1, 27) = 14.0$, $p < .001$, and block, $F(5, 135) = 9.8$, $p < .001$, were significant, but there was no interaction of group and block, $F(3, 135) = 1.1$, $p > .05$. Both groups became more consistent in time parameterization across practice, with the ConFB group having considerably smaller time factor VEs than the No-ConFB group throughout the practice phase. The main effects of group, $F(1, 27) = 16.8$, and block, $F(5, 135) = 16.0$, $ps < .001$, were significant. There was no interaction of group and block, $F(5, 135) < 1$.

Summary. Both groups demonstrated similar performances with regard to the accuracy of the fundamental movement pattern (GMP) during practice, but continuous feedback led to decreased stability in the GMP as compared with the No-ConFB condition. In terms of parameterization measures, however, continuous feedback facilitated accuracy and stability in both ampli-

tude and time parameterization (though the effect on amplitude factor VE was only marginally significant).

Results: Retention

One day after the practice phase, all participants returned to perform a retention test without feedback. Task Versions A, B, and C were presented in randomized order (four trials per version).

Overall performance. RMSEs in retention are shown at the top of Table 1. Errors were generally larger than those at the end of practice, with both groups showing similar error scores. The group effect was not significant, $F(1, 28) < 1$.

Motor-program measures. With regard to residual RMSE, the measure of GMP accuracy, the No-ConFB group demonstrated almost no retention loss as compared with performance at the end of practice, whereas the ConFB group showed a clear performance decrement (see Figure 3, right panel), to the point that the ConFB group had greater errors in the retention test. This group effect was significant, $F(1, 28) = 4.3, p < .05$. Even though both groups showed similar residual RMSEs during practice, the ConFB condition was less effective for learning of the GMP than the No-ConFB condition, at least as evaluated by this retention test.

For residual RMSE-VE, our measure of motor-program stability (Figure 4, right panel), the ConFB group was more variable in the fundamental movement pattern than the No-ConFB group. This effect was reliable, $F(1, 28) = 6.7, p < .05$. In addition to degrading the accuracy of the GMP, continuous feedback also degraded its stability, compared with the No-ConFB condition.

Parameterization measures. For accuracy of amplitude parameterization, both groups showed a performance decrement relative to performance at the end of practice (see Table 1). However, contrary to the situation in the practice phase, the ConFB group now had numerically larger errors than the No-ConFB group, but this difference was not significant, $F(1, 28) < 1$. For the variable errors in amplitude parameterization, the ConFB and No-ConFB groups showed

very similar values (see Table 1), and the effect of group was not significant, $F(1, 28) < 1$.

For time factor measures, the two groups tended to produce very similar ICEs in time parameterization (see Table 1), and this group effect was not reliable, $F(1, 28) = 1.9, p > .05$. However, even though the ConFB participants were more stable in time parameterization than the No-ConFB participants during practice, they were less stable (greater time-factor VEs) in retention (see Table 1). This group effect in retention was significant, $F(1, 28) = 6.6, p < .05$.

Summary. For retention performance, measures of overall accuracy showed no effect of concurrent feedback during practice. However, measures of GMP proficiency indicated that concurrent feedback during practice degraded the learning of both the accuracy and stability of the motor-program representation. The measures of parameterization accuracy and stability in retention were generally degraded by concurrent feedback during practice, but these effects were reliable only for the time factor variability measure. Thus the chief effect of concurrent feedback seemed to be a degrading influence on motor-program accuracy and stability and on time parameterization stability.

EXPERIMENT 2

Even though Experiment 1 was fairly clear in demonstrating degrading effects of concurrent feedback on program and parameter learning, one concern needed to be defused in a second experiment. We were concerned because in the ConFB condition, the participant saw the template over which his or her action was drawn in addition to the feedback, whereas the No-ConFB group received neither the feedback nor the template. It was therefore possible that the results we obtained could have been attributable to the inclusion of the template rather than to the addition of the feedback. Therefore, to hold the factor of the template constant, in Experiment 2 both ConFB and No-ConFB groups were shown the template during movement execution.

TABLE 2

Mean RMSE, Amplitude Factor |CE|, Amplitude Factor VE, Time Factor |CE|, and Time Factor VE in Practice and Retention in Experiment 2

	Practice Blocks						Retention
	1	2	3	4	5	6	
RMSE (overall performance)							
ConFB	241	212	177	167	162	162	322
No-ConFB	303	239	242	233	224	226	283
Amplitude Factor CE (amplitude parameter accuracy)							
ConFB	.08	.13	.10	.09	.07	.09	.23
No-ConFB	.17	.13	.14	.15	.12	.12	.23
Amplitude Factor VE (amplitude parameter stability)							
ConFB	.24	.22	.21	.19	.17	.19	.25
No-ConFB	.24	.20	.21	.21	.22	.23	.20
Time Factor CE (time parameter accuracy)							
ConFB	.08	.05	.04	.05	.03	.04	.42
No-ConFB	.11	.07	.05	.06	.05	.07	.23
Time Factor VE (time parameter stability)							
ConFB	.11	.08	.06	.08	.06	.08	.23
No-ConFB	.21	.14	.15	.11	.10	.10	.15

Method

Participants. Students from the University of Munich ($N = 32$) participated in exchange for DM 15 (about \$9.00 U.S.). They had not served in Experiment 1 and had no prior experience with the task.

Apparatus, task, and procedure. For Experiment 2, the apparatus, task, and procedures were identical to those for Experiment 1, except that during movement execution, the template was displayed not only for the ConFB group, as in the previous experiment, but also for the No-ConFB group. Participants were randomly assigned to one of the two conditions (16 per group).

Results: Practice

Overall performance. As can be seen from Table 2, both groups became more accurate across practice. As in Experiment 1, overall the ConFB group demonstrated more effective performance than the No-ConFB group. The main effects of block, $F(5, 150) = 12.7$, $p < .001$, and group, $F(1, 30) = 7.2$, $p < .05$, were significant. There was no interaction of group and block, $F(5, 150) < 1$.

Motor-program measures. Again, similar to Experiment 1, the No-ConFB and the ConFB groups

showed similar residual RMSEs during practice (see Figure 5). The main effect of group was not significant, $F(1, 30) < 1$. Only the block main effect was significant, $F(5, 150) = 5.3$, $p < .001$, indicating that both groups became more accurate with regard to the fundamental movement pattern, or GMP, across practice. The interaction of group and block was not significant, $F(5, 150) = 1.3$, $p > .05$.

Both groups also became more consistent in producing the fundamental pattern across practice blocks (see Figure 6). The main effect of block was significant for residual RMSE-VE, $F(5, 150) = 8.9$, $p < .001$. However, even though the ConFB group tended to be more variable than the No-ConFB group, the group main effect was not significant, $F(1, 30) < 1$. Also, the interaction of group and block was not significant, $F(5, 150) < 1$.

Parameterization measures. Amplitude factor |CE|s can be seen in Table 2. There was no clear error reduction across practice, and the main effect of block was not significant, $F(5, 150) < 1$. The ConFB group tended to be more accurate in amplitude parameterization than the No-ConFB group, but, contrary to Experiment 1, the group effect was not significant here, $F(1, 30) = 3.0$,

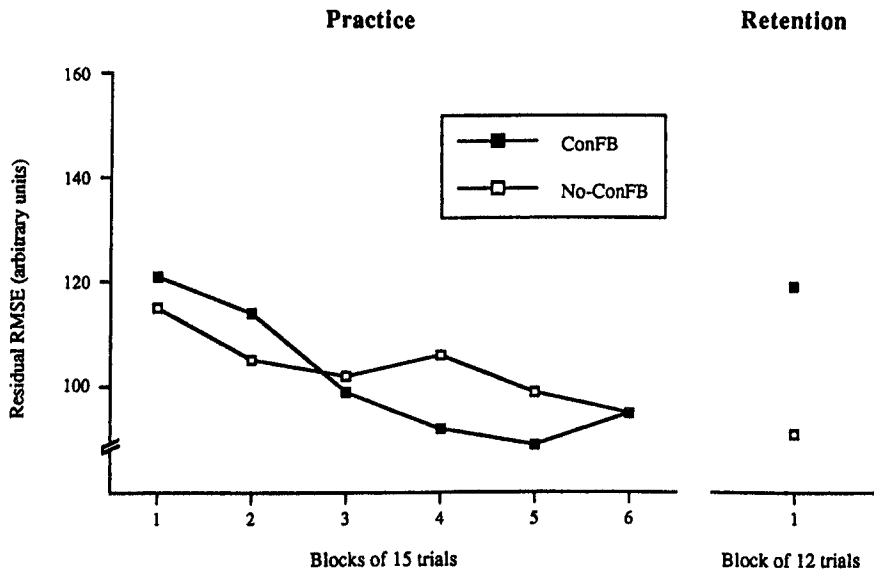


Figure 5. Residual RMSE in practice and retention in Experiment 2.

$p = .09$. There was no Group $\times$ Block interaction, $F(5, 150) = 1.2$, $p > .05$. In terms of the variability in amplitude parameterization, both groups tended to reduce their amplitude factor VEs across the practice phase (see Table 2), yet the main effect of block was only marginally significant,

$F(5, 150) = 2.2$, $p = .06$. The main effect of group, $F(1, 30) > 1$, and the Group $\times$ Block interaction, $F(5, 150) = 1.5$, $p > .05$, were not significant.

In time parameterization, the No-ConFB group tended to have lower |CE|s than the ConFB group

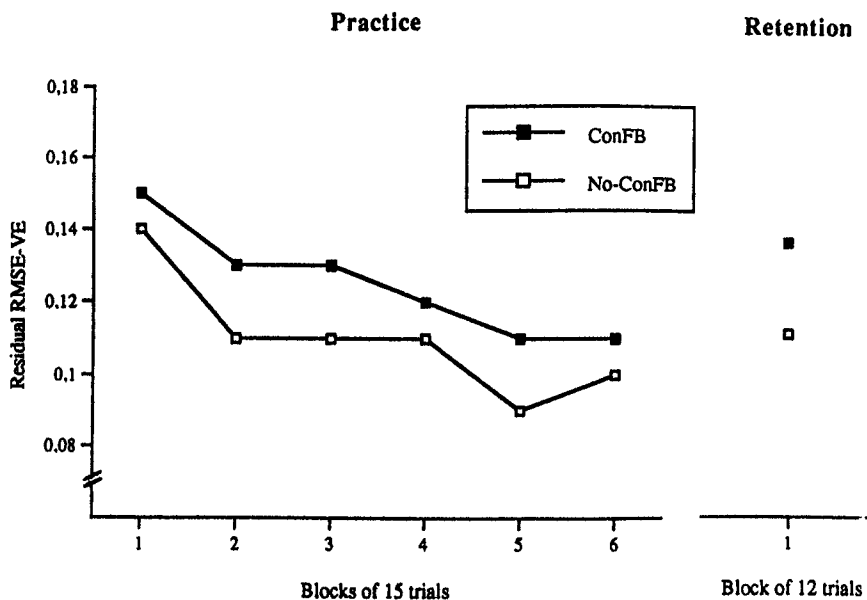


Figure 6. Residual RMSE-VE in practice and retention in Experiment 2.

(see Table 2). However, the main effect of group just failed to reach significance, $F(1, 30) = 3.3, p = .08$. Both groups became more accurate in time parameterization across practice blocks, $F(5, 150) = 4.0, p < .01$. The interaction of group and block was not significant, $F(5, 150) < 1$.

As in Experiment 1, the ConFB group was clearly more stable in time parameterization than the No-ConFB group (see Table 2). The group main effect, $F(1, 30) = 10.4, p < .01$, was significant. Also, the main effect of block was significant, indicating that both groups reduced their time factor VEs across practice, $F(5, 150) = 8.3, p < .001$. However, the No-ConFB group had clearly larger VEs than the ConFB group at the beginning of practice and showed a greater improvement across practice blocks. The Group $\times$ Block interaction was significant, $F(5, 150) = 2.4, p < .05$.

Summary. As in Experiment 1, the ConFB condition facilitated overall performance (RMSE) compared with the No-ConFB group; also, both groups demonstrated similar performance with regard to the fundamental movement pattern, or GMP, during practice. Concurrent feedback (ConFB) also tended to make GMP production more variable than postresponse feedback (No-ConFB), but this effect was not significant in the present experiment. In terms of parameterization, concurrent feedback again clearly facilitated stability in time parameterization, compared with the No-ConFB condition.

Results: Retention

Overall performance. As occurred in Experiment 1, even though the ConFB group was more effective than the No-ConFB group during practice, there were no significant group differences in RMSE in retention, $F(1, 30) = 1.6, p > .05$.

Motor-program measures. As to the accuracy of the GMP, the No-ConFB group demonstrated clearly lower residual RMSEs than the ConFB group (see Figure 4). The group effect was significant, $F(1, 30) = 7.7, p < .05$. Thus as in Experiment 1, concurrent feedback enhanced the learning of an accurate GMP. The ConFB group also tended to be more variable in the production of

the fundamental movement pattern (see Figure 5). However, the main effect of group failed to reach significance for residual RMSE-VE in this case, $F(1, 30) = 2.0, p > .05$.

Parameterization measures. Similar to Experiment 1, there were no group differences in terms of amplitude factor |CE|, $F(1, 30) < 1$, and VE, $F(1, 30) = 2.7, p > .05$. With regard to time factor |CE|, the ConFB group was less effective than the No-ConFB group. The group effect was significant, $F(1, 30) = 8.4, p < .01$. Also, as in Experiment 1, the ConFB group had clearly larger VEs in time parameterization than the No-ConFB group. This effect was significant, $F(1, 30) = 4.5, p < .05$.

Summary. The ConFB condition again degraded the learning of an accurate fundamental movement pattern (GMP), compared with the No-ConFB condition. In addition, stability in time parameterization was learned less effectively with the provision of concurrent feedback during practice (ConFB). These results were similar to those of Experiment 1 and indicate that the potential confound of having the template absent in the No-ConFB group was not a factor in the outcome.

DISCUSSION

We attempted to gain further insight into the functions of continuous, concurrent feedback for motor skill learning. Many investigators have found that concurrent feedback usually facilitates performance when it is present during practice but that it is detrimental to performance when measured in retention tests (or sometimes transfer tests). In this sense, concurrent feedback generally degrades learning. However, to our knowledge, there have been no previous experimental investigations of concurrent feedback effects in actions governed by primarily open-loop (or motor-program) control, and adding such information on this critical class of tasks seemed essential for a full understanding of the effect of feedback on learning. However, we also sought to examine the effects of concurrent feedback in our tasks and paradigm, such that the effects on motor-program learning could be separated from

the effects on parameterization learning (e.g., Wulf & Schmidt, 1994, 1997; Wulf et al., 1993).

The two experiments presented here differed with respect to only one methodological detail: Whereas in Experiment 1 the No-ConFB group did not see the template during movement production in practice (as did the ConFB condition), in Experiment 2 both groups could see the template while they were executing the movement (to make sure that any obtained effects were attributable to the addition of concurrent feedback, not to the inclusion of the template). The results of both experiments were relatively consistent, indicating that the effects we found for Experiment 1 resulted mainly from the provision of concurrent feedback.

Feedback, Programming, and Parameterization

In both experiments, from the global performance measure (RMSE) it was obvious that ConFB condition facilitated performance during practice, as compared with the No-ConFB condition. However, overall learning, as measured by performances in retention, tended to be degraded by the concurrent feedback, though these effects were not reliable. There were certainly no tendencies toward facilitation of learning by the concurrent feedback condition. But when the various components of error (associated with proficiency in programming versus proficiency in parameterization) were separated mathematically, several clear effects emerged.

Programming. In terms of the accuracy in motor programming, indexed here by residual RMSE, continuous feedback had essentially no effect during the practice phase. However, the effects on residual RMSE-VE tended to be detrimental (this effect was significant only in Experiment 1, though), in that concurrent feedback seemed to decrease the stability of the GMP representations. On measures of learning (seen in the retention test), concurrent feedback during practice generally degraded performance in terms of the accuracy of the GMP in both experiments. Also, with regard to the stability of the GMP, the ConFB group showed less effective learning than the No-ConFB group (even though

this effect was again significant only in Experiment 1). We interpret these findings as showing that concurrent feedback was detrimental for the learning of the motor-program representation, particularly with respect to its accuracy, and somewhat less clearly with regard to its stability.

Parameterization. For parameterization performance, a different picture emerged. During practice, concurrent feedback generally facilitated performance of most of the measures of parameterization. These effects began very early in practice and generally persisted throughout the practice phase (exception: amplitude factor VEs; see Tables 1 and 2). However, in the retention tests these trends were reversed, with the ConFB group usually showing poorer retention performance than the No-ConFB group. With the exception of time factor VE, these detrimental effects were varied; however, they were not always statistically reliable and were sometimes essentially absent numerically (e.g., amplitude factor VE).

Interestingly, with regard to variability in time parameterization (time factor VE), the group differences were consistent and significant in both experiments. Whereas the ConFB group was clearly less variable in absolute timing than the No-ConFB group during practice, these differences were reversed in retention. That is, concurrent feedback interfered with learning to time the whole action consistently. Overall, there was no tendency for concurrent feedback to benefit any of these measures of learning, and our interpretation is that this form of feedback is probably detrimental—and certainly is not beneficial—for parameterization learning.

Mechanisms of Continuous Feedback

These findings extend the existing work on concurrent feedback in several ways. First, most earlier studies have used tasks with long-duration movements (tracking, positioning, etc.) for which feedback processes tend to dominate response production. The present findings generalize these principles to more rapid actions, in which open-loop programming processes are more strongly involved. Our tasks were not extremely rapid and

ballistic (movement times of 937 ms), however, so it is difficult to be confident about the exact contributions of open- and closed-loop processes in these actions. The duration of the task is only one of the factors that determine the extent to which actions are structured in advance, however, as feedback processes can in some cases be far shorter than this 900-ms value. Tasks performed in closed, stable, and predictable environments generally tend to be structured in advance (clutch-accelerator movements in a car, operating a doorknob to open a door, etc.), which tends to free attentional processes from the analysis of feedback information for the detection of errors (Schmidt, 1988). It seems reasonable to suggest that, in any case, programming processes were considerably more involved in our task than they were in the earlier experiments on this question.

To the extent that programming and open-loop processes are involved in the present task, then, these data suggest that concurrent feedback degrades such processes during learning. Prior to this the detrimental effects of concurrent feedback (using mainly tasks with closed-loop control) had been explained by the general idea that concurrent feedback attracted attention and processing during performance, which interfered with the processing of relevant (i.e., intrinsic) sources of feedback. This generally facilitates performance temporarily, especially if the concurrent feedback provides information about movement accuracy, but it degrades learning, as seen in tests in which concurrent feedback has been removed. This is the most common finding when augmented concurrent feedback is used in simulators or other real-world learning settings in attempts to facilitate the learning of the tasks with the concurrent feedback removed in the transfer test.

Hypotheses that explain these results in terms of the substitution among sensory information sources do not account for our data very well, however, for several reasons. First, the pattern of results seen here appears to involve learning decrements in the open-loop (programmed) aspects of the action, which are presumably not dominated by feedback during performance. Also, our

measures of GMP proficiency were generally not facilitated by concurrent feedback during practice (as many closed-loop tasks seem to be) but were degraded only when measures were taken on retention tests. Such results present a challenge to understanding how the learning of open-loop responses, involving processes believed to be largely free from feedback influences, can be degraded by manipulations of the within-response sensory information.

A second reason that such a sensory substitution hypothesis appears to be inadequate is that our measures of parameterization accuracy were generally facilitated by concurrent feedback during practice, but the long-term effects on retention were spotty and unreliable (with the exception of stability in time parameterization, which was clearly degraded by concurrent feedback). Our measures of parameterization accuracy might be influenced by closed-loop processes, however, in which errors in the initially programmed parameterization could be adjusted easily when the learner sees that the movement is too fast (or slow) or too small (or large). If so, then one would expect the learning of such parameterization processes to be degraded by concurrent feedback, which was not the case here. There were tendencies in this direction, but the effects were not strong, reliable, or consistent (again, with the exception of time factor VE). However, there were no indications that concurrent feedback facilitated parameterization.

Perhaps the reason for these patterns of effects with this task can be thought of in terms of what can be called an *incomplete programming hypothesis*. According to this general view, prior to each performance, participants presumably must retrieve the GMP from memory and parameterize it so that it meets the specified amplitude and timing demands. However, when concurrent feedback is present during practice—and when the participant has learned that such feedback is generally effective for performance—these programming processes might be less completely or effectively done than would be the case if the participant did not expect such augmented feedback. Programming could be considered less complete in several ways.

One way that programming could be less complete is based on the idea that some actions are thought to be composed of separate, sequentially organized units of control (e.g., Schneider & Schmidt, 1995; Young & Schmidt, 1990). If so, the initial segments of the action could be pre-programmed, with later segments of the action left unprogrammed. During practice, concurrent feedback during the final portions of the action could allow one to practice the last segments without preprogramming them. If augmented concurrent feedback is provided in practice, such information could induce a greater tendency to use such a mode of control. The disadvantage is that if concurrent feedback is removed in retention, there would no longer be a learned basis for control of the last part of the action, and performance would suffer. Such a hypothesis might be tested in future work by examining the locus of decrements in the action (initial vs. final segments) when augmented feedback is withdrawn in retention or transfer tests.

A second way that incomplete programming could be conceptualized is that the whole action might be programmed—but less completely so, in some sense—prior to response initiation. The general plan of the entire action could be specified, perhaps as Greene (1972) has suggested, with his notion of the “ballpark response,” in which programming is sufficient only to bring the action “into the ballpark” in terms of its precision. However, when the response is then later generated, the precise details are added based on the feedback provided during the action. If so, then augmented concurrent feedback during practice could provide an increased reliance on the later, feedback-based control processes, and require less effort in terms of generating the precise details of control before the action is initiated. Again, such a mode of practice would benefit performance in acquisition, but it would be inadequate if the retention or transfer tests do not have the augmented feedback to provide the basis for adding fine control. These two subhypotheses are not exhaustive, and other analogous accounts could probably be formulated to account for the general pattern of results.

Training and Simulation

Whatever mechanisms are involved, our results join many others from a variety of tasks and paradigms in showing that when learning is measured on delayed retention and transfer tests, continuous concurrent augmented feedback is detrimental for learning. There seems to be little support for using these kinds of procedures in training, and in fact there is much support for the notion that such concurrent feedback should be avoided. This general notion has not, in our view, been readily accepted by trainers in many real-world training situations, considering that many such programs continue to promote continuous feedback as a way to facilitate performance and, ultimately, learning.

Probably one reason that the training difficulties with concurrent feedback have not been well accepted is because such feedback is usually a powerful variable for facilitating performance while it is being presented. Trainers are well aware of its effects, and it is understandable that well-motivated trainers would want to use whatever means they can to facilitate performance in practice. Furthermore, it seems natural to assume that any variable that is so effective for performance would also have powerful learning effects.

However, as we (Schmidt & Bjork, 1992) have pointed out with several experimental variables (including the use of various feedback manipulations), this assumption is not necessarily correct, and in many cases it appears to be wrong. Indeed, for several variables, deliberately organizing practice so that performance is “difficult” appears to provide decrements in performance during practice but sizable gains in performance on tests of learning conducted later. Assessing the effectiveness of variables operating in training appears to require a careful examination of the effects seen on long-term retention and/or transfer tests.

However, there are a few troubling exceptions to this generalization (e.g., Lintern, 1980; Swinnen et al., in press) that indicate that augmented concurrent feedback could be effective for learning

after all. In most demonstrations of the degrading effects of concurrent feedback on learning, the training tasks have been relatively simple laboratory activities (though this is not universally the case). One suggestion, therefore, is that task complexity might somehow prove to be a moderating variable in these situations, although more evidence is needed to be certain. However, for the majority of tasks reported in the human factors literature, concurrent feedback apparently degrades the learning of processes leading to the development of effective retention and/or transfer performance, which is critical to the success of real-world training and simulation programs.

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